

What is Soil

- ***Soil* is the thin layer of organic matter and inorganic matter that covers the earth crust and support life. It is a weathered layer of earth's surface .**

- Soil has four important functions:
 - as a medium for plant growth.
 - as a means of water storage,
 - water supply and
 - purification.

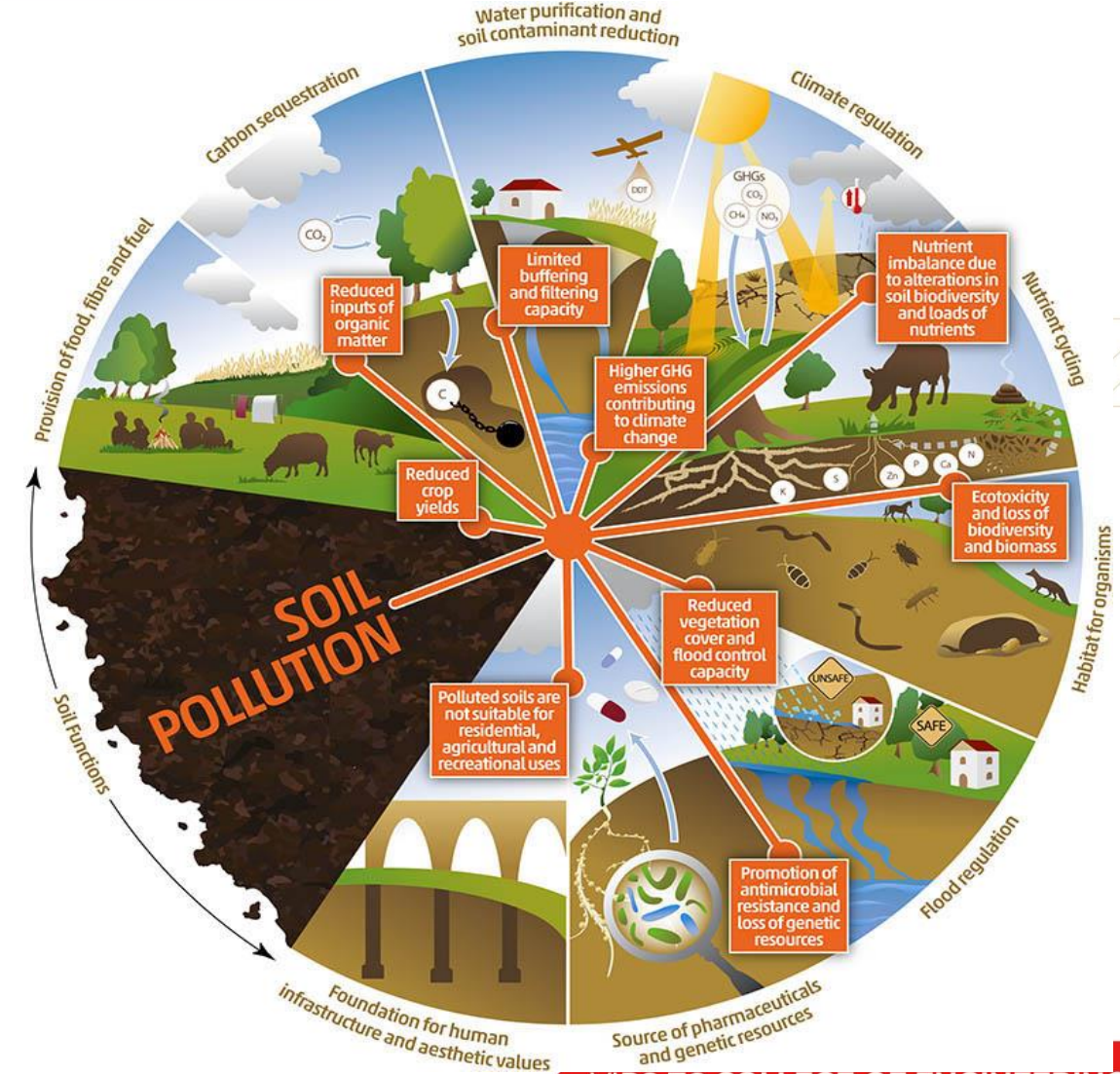


- Soil is composed of five major components
 - a. **Mineral matter** obtained by the distintergration and decomposition of rocks;
 - b. **Organic matter**, obtained by the decay of plant residues, animal remains and microbial tissues;
 - c. **Water**, obtained from the atmosphere and the reactions in soil (chemical, physical and microbial);
 - d. **Air or gases**, from atmosphere, reactions of roots, microbes and chemicals in the soil
 - e. **Organisms**, both big (worms, insects) and small (microbes)
- The typical soil consists of approximately 45% mineral, 5% organic matter, 20-30% water, and 20-30% air.

Soil Pollution

❑ . Soil Pollution is gradually becoming a major challenge that we need to overcome to establish a healthy environment. Soil contamination occurs either because of human activities or because of natural phenomena.

❑ **Soil pollution** is defined as the presence of toxic chemicals (pollutants or contaminants) in the soil, in very high concentrations to pose a risk to human health and the ecosystem. For example, exposure to soil containing high concentrations of benzene can increase the risk of soil pollution diseases like contracting leukaemia.



Sources of Soil Pollution

Industrial Pollution: The discharge of industrial waste into soils can result in soil pollution. In India, as mining and manufacturing activities are increasing rapidly, soil degradation is also increasing. The extraction of minerals from the earth is responsible for affecting soil fertility. Whether it is iron ore or [coal](#), the by-products are contaminated, and they are disposed of in a manner that is not considered safe. As a result, the industrial waste stays on the soil surface for a long duration and makes it unsuitable for further use.

Agricultural Activities: The use of insecticides and pesticides for a long period can cause soil pollution. Repetitive use can cause insects and pests to become resistant to it. Instead of killing pests and insects, it degrades the soil quality. They are full of chemicals that are not produced in nature and cannot be broken down by them. As a result, they seep into the ground after they mix with water and slowly reduce the fertility of the soil. Plants absorb many of these pesticides, and after decomposition cause soil pollution.

Waste Disposal: Disposal of plastics and other solid waste is a serious issue that causes soil pollution, disposal of electrical items such as batteries causes an adverse effect on the soil due to the presence of harmful chemicals. Eg: [lithium](#) present in batteries can cause the leaching of soil. Human waste such as urine, faeces, diapers, etc is dumped directly in the land. It causes both soil and water pollution.

Sources of soil pollution....

Acid Rain: It is caused when pollutants present in the air mix with the rain and fall back on the ground. The polluted water could dissolve away some of the essential nutrients found in soil and change the structure of the soil thus making it unsuitable for agriculture.

Heavy Metals: The presence of heavy metals (such as lead and [mercury](#)) in very high concentrations present in soils can cause them to become highly toxic for human beings.

Nuclear Waste: It can also lead to soil degradation.

Oil Spills: Oil leaks can happen during the storage or transport of chemicals, the chemicals present in the fuel deteriorates the quality of soil and make them unsuitable for further cultivation, chemicals can also enter into the groundwater through the soil, and hence it will make water undrinkable.

Effects of Soil Pollution

1. **Effect on Health of Humans** : The contamination of soil has major consequences on our health. Crops and plants grown on polluted soil absorb the toxicant and then pass these to consumer.
2. **Effect on Growth of Plants** : Most plants are unable to adapt when the chemistry of the soil changes so radically. Fungi and bacteria found in the soil that bind it together begin to decline, creating an additional problem of soil erosion making land unsuitable for agriculture and any local vegetation to survive.
3. **Decreased Soil Fertility** : The toxic chemicals present in the soil can decrease soil fertility and therefore decrease in the soil yield. The contaminated soil is then used to produce fruits and vegetables which lack quality nutrients.
4. **Toxic Dust** : The emission of toxic and foul gases from landfills pollutes the environment and causes serious effects on health of people also.
5. **Changes in Soil Structure** : The death of soil organisms e.g. earthworms in the soil can lead to alteration in soil structure. Apart from that, it could also force other primary consumer species to move to other places in search of food leading to ecosystem imbalance.

Various measures to control soil pollution are:

- **Discouraging open dumping .**
- **Banning of synthetic pesticides and decrease usage of inorganic fertilizers .Instead biopesticides should be preferred .**
- **Following crop rotation to revive depleted nutrients and plantation of more trees .**
- **High concentration of heavy metals can be controlled by growing special grass by the name of vetiver grass . The contaminants concentrates on its root and reduce amount of Pb and Cd in soil ..**
- **Organic methods of farming are being supported to curb menace of soil pollution**
 - **Pneumatic pipes should be laid for collecting and disposing of wastes.**
 - **Materials like paper, glass, and plastics can be recycled.**
 - **Metals should be recovered from scrap and disposed of materials.**

Thermal Pollution

- Different industries and nuclear power plants use water for cooling purposes and discharge the heated water into nearby streams or water bodies. As a result, the nearby water body is heated up and this in turn, affects aquatic life. This phenomenon is known as thermal pollution.
- It is caused by:
- Natural phenomenon such as earthquakes also cause thermal pollution.

Effects of thermal pollution

The harmful effects of thermal pollution are discussed below:

1. Reduction in dissolved Oxygen

The pollutant from various industrial plants are heated decreases the concentration of oxygen with an increase in the temperature of water.

2. Change in water properties

The decrease in density, viscosity, and solubility of gases in water increases the setting speed of suspended particles which seriously affect the food supplies of aquatic organism.

3. Increase in toxicity

The concentrated pollutant causes the rise in the temperature of water which increases the toxicity of the poison present in water. The toxicity in water will increase the death rate in marine life.

Control measures

- Thermal pollution can be controlled by passing the heated water through a cooling pond or a cooling tower after it leaves the condenser.
- To construct a large shallow pond. Hot water is pumped into one end of the pond and cooler water is removed from the other end

- Radiation pollution is caused by the addition of more ionizing radiation to the environment and people are exposed to more radiation than they normally experience. Radiation interacts with living tissues and damages them.

The sources of radiation may be:

- Natural
- Artificial(man-made)

Natural Sources of Radiation

- **Cosmic rays** are high-energy ionizing electromagnetic radiation. The cosmic rays originate from the stars in our galaxy by virtue of nuclear reactions primarily in their cores. The cosmic rays are constantly reaching the earth from outer space.
- **Naturally occurring radioisotopes** such as radon-222 found in soil in small quantity is another source of radioactive radiations.
- **Radioactive elements** which like uranium, thorium, radium, isotopes of potassium (K-40), and carbon (C-14) occur in the lithosphere. Potassium-40 contributes radioactivity to all potassium-containing systems in the soil. Crops grown on such soil contain radioactive elements like carbon-14 and potassium-40. Water gets contaminated with various radionuclides when it runs through soils and rocks containing radioactive minerals.

- **Diagnostic medical applications:** Radiations are employed for diagnostic and therapeutic applications. X-rays are used in general radiology and CT scan. Gamma rays are used in treatment of cancer. In all these procedures we are exposed to varying doses of radiations.
- **Nuclear Tests:** Nuclear explosion tests especially when carried out in the atmosphere are a major cause of radiation pollution. It is responsible for increasing the background level of radiation throughout the world
- **Nuclear Reactors:** Radiations may leak from nuclear reactors and other nuclear facilities even when they are operating normally. It is often feared that even with the best design, proper handling and techniques, some radioactivity is routinely released into the air and water.

Following measures can help in controlling the radioactive pollution:

- Workers in nuclear plants should be provided with nuclear gadgets and safety measures against accidents.
- Leakage of radioactive elements from nuclear reactors, laboratories, transport, careless handling and use of radioactive fuels should be checked.
- Level of radiation pollution should be monitored regularly in risk areas.
- Disposal of radioactive wastes deserves special attention.

Noise Pollution

- In addition to other pollutants, noise also plays a major role in the degradation of the environment. Noise can be explained as any sound that is not desired by recipient. So noise pollution is unwanted sound dumped into the atmosphere without regard to the adverse effects it may have.
- Our ears can hear ordinary conversation between 30-60 decibels. Modern conversation has a noise value of 60 decibels. A decibel value greater than 80 decibels causes noise pollution.



Permissible Sound Levels:

Type of area	Day Time	Night Time
Industrial area	75	70
Commercial area	65	55
Residential area	55	45
Silence zone	50	40

Effect of Noise Pollution

- Constant noise affects a man physically and mentally. Physical effects include blood vessels to contract, skin to become pale, muscles to constrict and rise in blood pressure leading to tension and nervousness.
- High intensity sound emitted by industrial plants, bottling machines, supersonic aircrafts, when continued for long periods of time not only disturbs but also permanently damages hearing.
- Offices, industries and crowded places where constant noise prevails can produce temper tantrums, headaches, fatigue and nausea
- Loud and sudden noise affect the brain. Intermittent noise leads higher incidence of psychiatric illness and also a danger to health of pregnant mothers and small infants.
- Noise has harmful effects on nonliving materials too, e.g. cracks develop under the stress of explosive sound

Following methods can control noise pollution:

1. Limited use of loudspeakers and amplifiers.
2. Exercising control over noise producing vehicles.
3. Industrial workers should be provided with ear plugs.
4. Delocalisation of noisy industries far away from dwelling units.
5. Within a radius of 10 miles of airport, no buildings or factories should be allowed.
6. Plants and trees should be planted all around the hospitals, libraries and schools and colleges